Exponential sums and the Riemann zeta function

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Abstract

Bombieri and Iwaniec's analogue for exponential sums of the Hardy-Littlewood method for exponential integrals is explained, with some further ideas from the author and N. Watt.

This paper is intended to set in context the joint work of Huxley and Watt [5]. Landau said in his inaugural dissertation, "You cannot draw a hard and fast line in mathematics between number theory and analysis". The fact that

$$\left|\int_1^2 G(x)dx - \frac{1}{M}\sum_{m=M}^{2M-1} G(\frac{m}{M})\right| \leq \frac{V}{M}$$

(where V is the total variation of the function G) is analysis. However if G is a complex exponential

$$G(x) = \exp 2\pi i T F(x) = e(TF(x)),$$

where $T > M$ and $F(x)$ is monotone, the integral is

$$\ll 1/T \min |F'(x)|$$

in Vinogradov's order of magnitude notation, whilst the sum may take any value between ± 1. Since the integral is numerically small, we ask whether the sum is small also.

If $F(x)$ is a polynomial of degree d with integer coefficients, and $T = M^d$, then $TF(m/M)$ is always an integer, and the sum is 1. To obtain a nontrivial bound, we must have the derivatives $F^{(r)}$ nonzero on the interval $[1,2]$ for $r \leq d+1$, so that $F(x)$ cannot be approximated by a polynomial of small degree. The simplest approach is to approximate by a polynomial of degree $d+1$, and then difference it, each step reducing both the degree and T.

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Van der Corput introduced a branching iteration: either to difference $F(x)$ or to apply Poisson summation to the sum. The more non-vanishing conditions that the derivatives satisfy, the deeper the iteration goes. The bounds for the sum take the form

$$\ll M^{-1+l-k}T^k,$$

where (k,l) is called an exponent-pair. The simplext non-trivial exponent-pair $(1/2,1/2)$ gives

$$\ll T^{1/2}/M.$$

For T very large we expect $e(TF(x))$ to behave like a random variable and the sum to be usually $1/M^{1/2}$ in size.

In number theory the Riemann zeta function [7,11] has

$$\zeta(\sigma - iT) \simeq \sum_{m=1}^{T} m^{iT-\sigma}, \qquad m^{iT} = e\left(\frac{T}{2\pi}\log m\right).$$

The sum behaves like

$$M^{iT-\sigma}\sum_{m=M}^{2M-1} e\left(\frac{T}{2\pi}\log\frac{m}{M}\right),$$

which is of the form considered above. Thus if (k,l) is an exponent-pair,

$$\zeta(l-k+iT) \ll T^k \log T.$$

Our knowledge of $\zeta(s)$ is least on the line $\operatorname{Re} s = 1/2$, so exponent-pairs with $l = k + 1/2$ are the most interesting. Rankin [10] found the limit of the iteration in this case to be $k = 0.1645\ldots$, $l = 0.6645\ldots$, and recently Graham [3] has described how to find the limit of the iteration in general, confirming Rankin's calculations. Van der Corput also applied his method to sums in two variables. The region of summation becomes two-dimensional, and boundary effects can dominate. Graham and Kolesnik are preparing a much-needed book on this [4]. Kolesnik reduced the Lindelöf exponent, the θ for which

$$\zeta(1/2+iT) \ll T^{\theta+\varepsilon}$$

for any $\varepsilon > 0$, from Rankin's $0.1645\ldots$ to $0.1620\ldots$.

Bombieri and Iwaniec [1,2] have reduced the Lindelöf exponent to $9/56 = 0.1607\ldots$ by a new method. The account given here of their ideas evolved form a three semester study seminar on preprint versions of [1] and [2] and

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will be fully written up by the author and Nigel Watt. It is an analogue for sums of the Hardy-Littlewood method for exponential integrals [12].

Step 1. Major and minor arcs. Divide the range M to $2M$ into subintervals of length N. All derivatives of $f(x) = TF(x/M)$ are monotone, choose an integer m in each subinterval so that $f''(m)/2$ is close to a rational a/q. Expand $f(m+n)$ as a power series in n up to the cubic term. Replace the coefficient of n by a rational number b/qs. The coefficient of n^3 is some number $\mu < 1$, the constant term doesn't affect the modulus. Major arcs have q small, minor arcs gave $q \geq R$, a parameter of the method.

Step 2. Gauss sums. The terms in n and n^2 give a Gauss sum modulo qs, which can be expressed in terms of complete Gauss sums. This is really a finite Poisson summation. The method stops here on the major arcs; if all arcs are treated as major, we recover the exponent-pair $(1/6, 2/3)$.

Step 3. Poisson summation over n. The sum over n of length N arising from a minor arc becomes a sum of length about H in a new variable h. The terms in n^2 and n^3 give terms in h^2, $h^{3/2}$ and $h^{1/2}$ and a summand

$$e(\alpha h^2 + \beta h + \gamma h^{3/2} + \delta h^{1/2}) = e(\mathbf{x} \cdot \mathbf{y})$$

with α and β rational,

$$\mathbf{x} = \mathbf{x}(h) = (h^2, h, h^{3/2}, h^{1/2}), \qquad \mathbf{y} = (\alpha, \beta, \gamma, \delta).$$

The coordinates α, β, γ, δ depend on the minor arc.

Step 4. The large sieve. In one form of the Hardy-Littlewood method Vinogradov's Mean Value Theorem is used to show that most of the minor arc sums are small. It is replaced by a form of the Large Sieve (hitherto unpublished?) in which both the "points" $\mathbf{y}$ and the "integer vectors" $\mathbf{x}$ may coincide. As the vectors $\mathbf{x}(h)$ form a thin set, consider

$$\mathbf{x}(h_1, h_2, h_3, h_4) = \mathbf{x}(h_1) + \mathbf{x}(h_2) + \mathbf{x}(h_3) + \mathbf{x}(h_4),$$

which corresponds to raising the sum to the fourth power.

Step 5. Semicubical powers of integers. This step is analogous to Vinogradov's Mean Value Theorem. "Integer points" $\mathbf{x}$ coincide if

$$h_1^2 + h_2^2 + h_3^2 + h_4^2 = k_1^2 + k_2^2 + k_3^2 + k_4^2, \tag{A1}$$

$$h_1 + h_2 + h_3 + h_4 = k_1 + k_2 + k_3 + k_4, \tag{A2}$$

$$h_1^{3/2} + h_2^{3/2} + h_3^{3/2} + h_4^{3/2} \approx k_1^{3/2} + k_2^{3/2} + k_3^{3/2} + k_4^{3/2}, \tag{A3}$$

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$$h_1^{1/2} + h_2^{1/2} + h_3^{1/2} + h_4^{1/2} \approx k_1^{1/2} + k_2^{1/2} + k_3^{1/2} + k_4^{1/2}. \qquad (A4)$$

Count the number of coincident pairs; Watt [13] has a nice method, using the fact that if $h_1, \dots, k_4$ satisfy (A1) and (A2), then so do $h_1 + t, \dots, k_4 + t$ for any t.

Step 6. Points indexed by the minor arcs. Points $\mathbf{y}$ coincide if

$$\left\| \overline{\frac{4a}{q}} - \overline{\frac{4a_1}{q_1}} \right\| \ll \frac{1}{H^2}, \qquad (B1)$$

$$\left\| \overline{\frac{2ac}{q}} - \overline{\frac{2a_1c_1}{q_1}} \right\| \ll \frac{1}{H}, \qquad (B2)$$

$$\frac{\mu_1 q_1^3}{\mu q^3} = 1 + O\left(\frac{R^2}{N^2}\right), \qquad (B3)$$

$$\left\| \frac{b}{s} - \frac{b_1}{s_1} \right\| \ll \frac{Q}{N}, \qquad (B4)$$

where N, R, μ, and H were introduced above, $c = [b/s]$ and Q is the size of q. Count the number of coincident points.

Step 6 is the most difficult. Bombieri and Iwaniec could only treat the case $F(x) = \log x$. Conditions (B2) and (B4) involve the first derivative of $f(x)$, which changes rapidly. Conditions (B1) and (B2) are twice as strong, so ignoring (B2) and (B4) loses only one third of the information. However one factor R^2/N^2 goes to compensate a loss in the Large Sieve step. If (B2) and (B4) could be used fully, one might get 3/20=0.15 for the Lindelöf exponent.

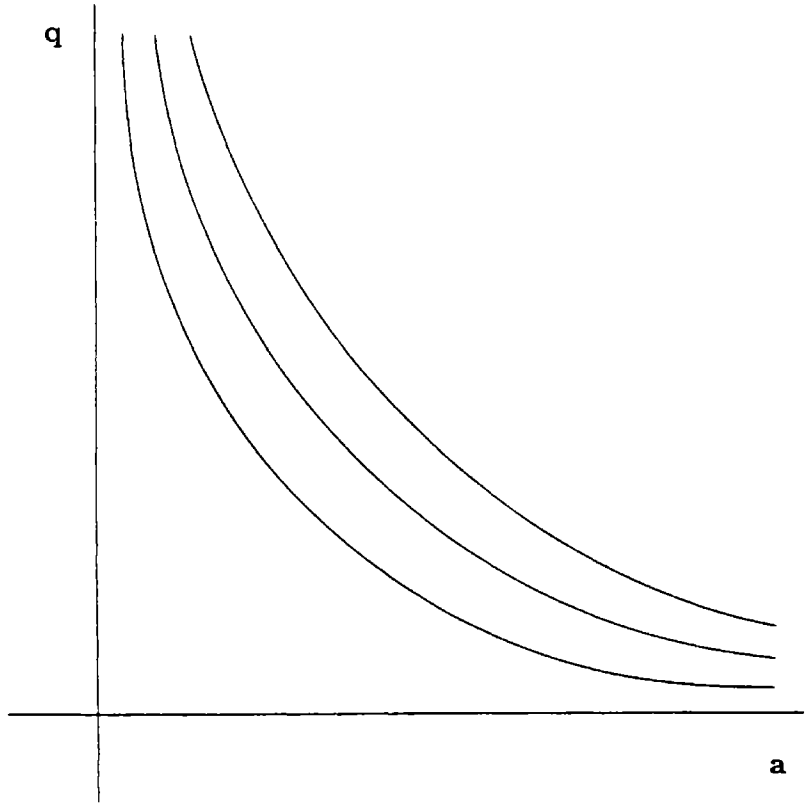
To tackle Step 6, consider the integer point $(4a, q)$ in two dimensions. Condition (B1) says

$$\begin{pmatrix} 4a_1 \\ q_1 \end{pmatrix} = M \begin{pmatrix} 4a \\ q \end{pmatrix}, \qquad M \in SL(2, \mathbf{Z}),$$

where M is a matrix with small entries. Condition (B3) says that $(4a, q)$ and $(4a_1, q_1)$ lie close to the same member of some given family of curves (Figure 1). For $F(x) = \log x$ the curves are hyperbolae $aq =$ constant; for $F(x) = x^{3/2}$ they are straight lines $a =$ constant. In these two cases we can

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use algebra.



At first we used Dirichlet's box principle in the general case: if there are many coincident pairs, there must be coincident pairs with some other common property (defining the "box" of the box principle). These were our intermediate results:

1. Boxes defined by $q = q_1$ (July 1986), leading to an exponent-pair $(2/13, 35/52)$.

2. Boxes defined by $q \simeq q_1$ (September 1986), leading to an exponent-pair (13/80,53/80).

3. Boxes defined by $q \simeq q_1$, arc of the curve replaced by a straight line (December 1986), exponent-pair (11/68,45/68).

Result 3 led to an iterative argument somewhat akin to finding an eigenvector of the matrix M, in which the box principle was no longer used, and the

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exponent-pair (9/56,37/56) corresponding to Bombieri and Iwaniec's bound for the Lindelöf exponent. Strictly one must add $\varepsilon \ll \log\log T/\log T$ to one member of the exponent-pair.

The new method is ill-suited to two-dimensional sums, as one needs simultaneous Diophantine approximation to all the partial derivatives of the function inside the exponential. The sum

$$\sum_h \sum_m e\left(TF\left(\frac{m+h}{M}\right) - TF\left(\frac{m}{M}\right)\right)$$

arises from the Weyl differencing step in Van der Corput's method. All the partial derivatives reduce to derivatives of $F(x)$, so the approximation is easier. Iwaniec and Mozzochi [6] treat the related but easier exponential sum

$$\sum_h \sum_m e\left(\frac{hT}{M}F'\left(\frac{m}{M}\right)\right)$$

with $F(x)$ again equal to $\log x$, obtaining new results for the divisor and circle problems. The method is modified as follows.

Step One: Major arcs are treated separately.

Step Two: Poisson summation over n on the minor arcs.

Step Three: Poisson summation over h on the minor arcs.

Step Four: A two-dimensional region of summation must be converted to a rectangle for the large sieve.

Step Five: More difficult coincidence conditions.

In Step 3 a stroke of mathematical luck involving the $J_{1/2}$ Bessel function prevents an error term catastrophe. Step 6 is unchanged. We hope to substitute the version of Step 6 outlined above to obtain the Iwaniec-Mozzochi circle problem result

$$AM^2 + O(M^{7/11+\varepsilon})$$

from [6] for the number of integer points inside a fixed smooth curve of area A enlarged by a factor M.

We would like to treat the difference of $F(x)$ in the same way. However Steps 3 and 5 go wrong, so we do not expect to be able to claim that (7/44,29/44) is an exponent-pair by the next CITN in Laval.

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